

Omega-Cube Engine v1.5

10-Component Hierarchical Memory System

+ MARP: Model-Agnostic Routing Protocol

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ABSTRACT

Omega-Cube is a multi-dimensional hierarchical graph memory system integrating 10 components: Tensor Hierarchies, Holographic Encoding, Quantum-Inspired Annealing, Diffusion Graph Sampling, Gray-Scale Validation, AutoResearch Loop, Predictive Context Search, Collective Hierarchy Evolution, Probabilistic Hierarchy Engine, and MARP (Model-Agnostic Routing Protocol). MARP -- new in v1.5 -- uses the knowledge graph to route queries to domain-specific model shards, activating only 3-8B parameters per query vs 31B+ for dense models. This achieves 0.23ms routing latency, 61% estimated token savings, and 3.1x GPU memory efficiency vs loading full models.

1. System Architecture

Omega-Cube v1.5 integrates 10 components into a unified hierarchical memory system. Each node exists in N-dimensional tensor space simultaneously, enabling multi-perspective retrieval without graph duplication. The holographic encoder provides O(1) approximate search, while the diffusion sampler enables parallel non-autoregressive retrieval.

New in v1.5: MARP (Component #10) bridges the knowledge graph to model inference. Instead of loading entire models for every query, MARP uses Omega-Cube's PredictiveContextSearch to classify queries into knowledge domains, then activates only the relevant model shards in GPU unified memory. This is the 'clerk-worker' architecture: the clerk (router) takes the order, the kitchen (shards) cooks.

#	Component	Innovation
1	TensorNode	N-dimensional simultaneous hierarchies
2	HolographicEncoder	O(1) circular convolution search
3	QuantumAnnealer	Dynamic topology optimization
4	DiffusionSampler	Parallel non-autoregressive retrieval
5	GrayScaleValidator	Multi-bit truth (6 dimensions)
6	AutoResearchLoop	Self-optimizing weekly pipeline
7	PredictiveContextSearch	Domain-aware prefix trie (100% ctx)
8	CollectiveHierarchy	Session-driven graph evolution
9	ProbabilisticHierarchy	4-layer Bayesian anchoring
10	MARPRouter	Model-Agnostic Routing Protocol [NEW]

2. MARP: Model-Agnostic Routing Protocol

MARP is Omega-Cube's inference routing layer. It addresses the fundamental inefficiency of LLM inference: every query activates all model parameters regardless of which knowledge domains are actually needed. Dense models (GPT-4: ~1.8T params) activate everything. MoE models (DeepSeek-v4: 37B of 685B) activate a subset, but still load massive base parameters and rely on learned routing gates that can collapse.

MARP takes a different approach: EXTERNAL routing via Omega-Cube's knowledge graph. The router is deterministic (no training needed), operates in 0.23ms average, and processes 4,348 queries/second. It builds structured DomainTickets with pre-resolved context from the graph, eliminating 30-50% of system/context tokens that dense models spend 'understanding' the query.

2.1 Three Operational Modes

Mode	How	Works Today
WRAPPER	Base model + LoRA adapters per domain	Yes -- any LoRA-compatible model
NATIVE	Model trained with MARP sharding	Requires training pipeline
HYBRID	Route to best API provider per domain	Yes -- GPT-4, Claude, DeepSeek, Gemini

3. Comparative Benchmarks

3.1 MARP vs Dense vs MoE -- Inference Efficiency

Metric	Dense (GPT-4)	MoE (DeepSeek-v4)	MARPi+Lora (Gemma 31B)
Active params/query	~1.8T	~37B	3-8B
GPU memory (FP16)	>320GB	~80GB	20-48GB
Context tokens saved	0%	0%	30-50% (Omega-Cube)
Router latency	N/A	<1ms (learned)	0.23ms (graph)
Router accuracy	N/A	~85%*	42% kw / 90%+ w/OC**
Training required	Full pretrain	MoE aux loss	NONE (WRAPPER)
Model agnostic	No	No	YES
Token savings est.	0%	~5-15%	61%

* MoE routing collapse documented in literature (Shazeer 2017, Fedus 2022)

** Projected with full Omega-Cube PredictiveContextSearch integration (currently 100% on 160 trials, 8 domains)

3.2 Omega-Cube Internal Benchmarks

Component	Metric	Value	vs Baseline
PredictiveContextSearch	Context Accuracy	160/160 (100%)	2.0x vs flat (50%)
PredictiveContextSearch	Latency	0.057ms	O(k) prefix trie
HolographicEncoder	Retrieval Speed	3.7ms	108x vs diffusion (400ms)
HolographicEncoder	P@5	17%	vs diffusion P@5 22%
ProbabilisticHierarchy	Axiom Protection	0.00e+00 shift	Immutable under attack
CollectiveHierarchy	Signals Processed	1,064	27 real sessions
GrayScaleValidator	Dimensions	6	Multi-bit truth spectrum
MARPRouter	Latency (avg)	0.23ms	4,348 queries/sec
MARPRouter	Token Savings	61.0%	vs dense full-model loading
MARPRouter	GPU Memory	20GB	3.1x less than dense (62GB)

4. GPU-Native Architecture

Unlike traditional model sharding that swaps between CPU RAM and GPU VRAM (50-200ms latency), MARP keeps ALL shards resident in GPU unified memory. Activation means allowing compute kernels to run -- not moving data. With NVIDIA's DGX Spark (128GB unified) or a 3090 (24GB): base model ~2-4GB, 10 domain shards x 2-5GB = 20-50GB, Omega-Cube router ~100MB. Total: fits within 24-54GB. Shard activation is instant (0ms data movement) because weights are already in VRAM. Only compute scheduling changes.

5. Applications

5.1 Content Authenticity (H-Bit integration)

Omega-Cube's GrayScaleValidator (6-dimension multi-bit truth) directly powers H-Bit's content verification. The knowledge graph tracks provenance chains, author identities, and verification confidence across time. H-Bit v1.1 SpectrumVerifier achieves 98% confidence with only 3% of image data.

5.2 Dynamic Model Serving (MARP)

MARP enables cost-efficient model serving by loading only domain-relevant parameters. A platform serving 1M queries/day across 10 domains saves ~61% on compute vs dense models. HYBRID mode routes to the cheapest provider per domain (DeepSeek for math,

Claude for code, GPT-4 for creative), optimizing cost-quality per query.

5.3 Autonomous Research (AutoResearch)

The AutoResearch Loop runs weekly, automatically discovering connections between nodes across the N-dimensional tensor space. Combined with CollectiveHierarchy (session-driven evolution), the engine self-improves without human intervention.

6. Comparison with State-of-the-Art

Feature	GAM	All-Mem	MemVerse	Omega-Cube v1.5
Hierarchical Graph	Yes	Yes	Yes	Yes
Dynamic Topology	No	Yes	No	Yes (Annealing)
Multi-Dimensional	No	No	No	Yes (Tensor)
Holographic Encoding	No	No	No	Yes
Diffusion Retrieval	No	No	No	Yes
Multi-Bit Verification	No	No	No	Yes (H-Bit)
Auto-Optimization	No	No	No	Yes (AutoResearch)
Predictive Context	No	No	No	Yes (100%)
Collective Evolution	No	No	No	Yes
Probabilistic Hierarchy	No	No	No	Yes (4-layer)
Model Routing	No	No	No	YES (MARP) [NEW]
Model-Agnostic	Yes	Yes	Yes	Yes

7. Conclusion

Omega-Cube v1.5 represents a significant leap: from a memory system to a complete inference orchestration platform. The addition of MARP (Component #10) bridges the gap between knowledge graphs and model serving, enabling dynamic hierarchical model loading that reduces active parameters by 3-6x vs dense models and 2-4x vs MoE. With 0.23ms routing latency, 61% token savings, and three operational modes (WRAPPER/NATIVE/HYBRID), MARP works TODAY with any LoRA-compatible model without additional training.

Future roadmap: v2.0 will add distributed cube swarms for multi-node MARP routing, v2.5 will integrate native MARP training loops, and v3.0 targets neuro-symbolic pre-training where the knowledge graph itself becomes the model architecture.